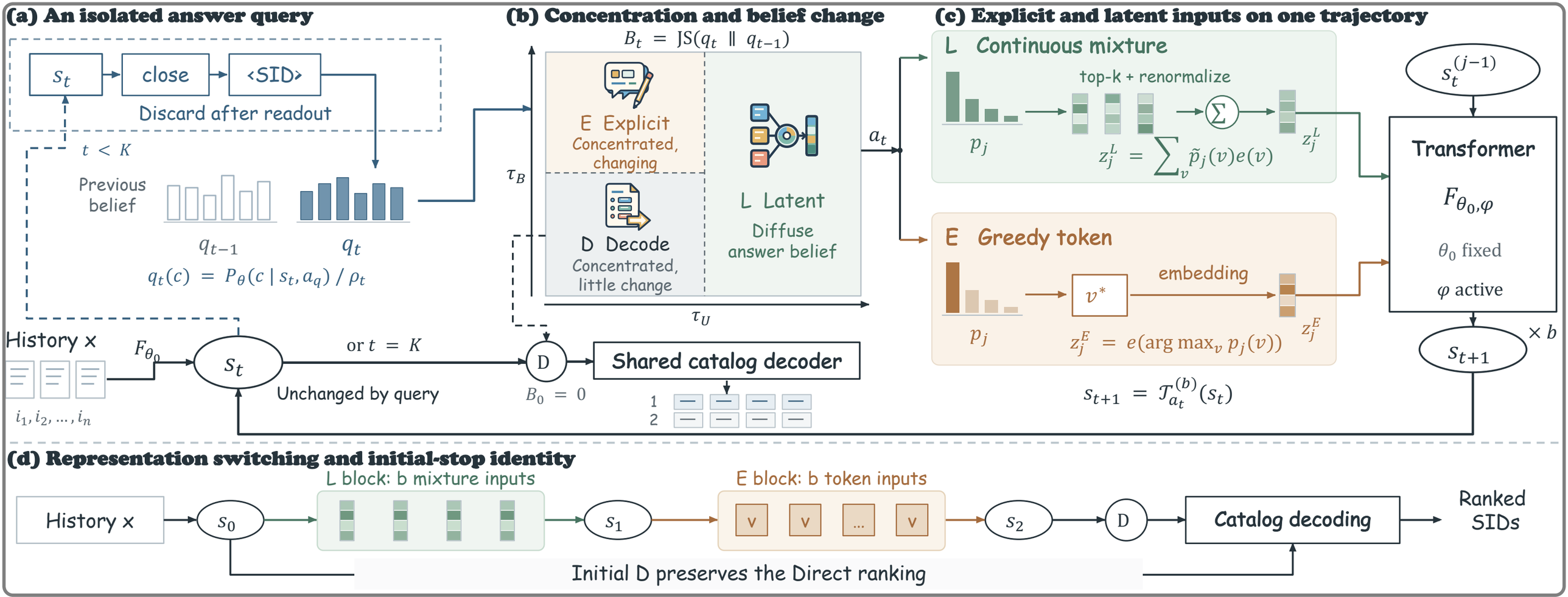

(a) An isolated answer query
s_t
close
<SID>
Discard after readout
$t < K$
Previous belief
q_{t-1}
q_t
$q_t(c) = P_\theta(c \mid s_t, a_q) / \rho_t$
History x
$i_1, i_2, \ldots, i_n$
F_{θ_0}
s_t
Unchanged by query
or $t = K$
(b) Concentration and belief change
$B_t = \mathrm{JS}(q_t \parallel q_{t-1})$
τ_B
E Explicit
Concentrated, changing
D Decode
Concentrated, little change
L Latent
Diffuse answer belief
τ_U
D
$B_0 = 0$
Shared catalog decoder
1
2
a_t
(c) Explicit and latent inputs on one trajectory
L Continuous mixture
top-k + renormalize
p_j
$z_j^L = \sum_v \tilde{p}_j(v) e(v)$
z_j^L
E Greedy token
p_j
v^*
embedding
$z_j^E = e(\arg\max_v p_j(v))$
z_j^E
$s_t^{(j-1)}$
Transformer
$F_{\theta_0,\varphi}$
θ_0 fixed
φ active
s_{t+1}
$\times b$
$s_{t+1} = \mathcal{T}_{a_t}^{(b)}(s_t)$
(d) Representation switching and initial-stop identity
History x
s_0
L block: b mixture inputs
s_1
E block: b token inputs
v
v
...
v
s_2
D
Catalog decoding
Ranked SIDs
Initial D preserves the Direct ranking